Recap – dealing with VLE

Binary mixtures have 3D phase diagrams

Reduced phase diagrams are 2D slices through 3D diagram

$$\gamma_i x_i f_i^l = \overline{f}_i^v = y_i f_i^v$$

$$\gamma_i x_i P_i^{SAT} = y_i P$$

Dissolved gases (above critical point):

problem is that standard state (pure liquid) does not exist

two approaches:

(a) extrapolate hypothetical pure liquid state beyond critical point

(b) define Henry's constant from slope of P-x plot at infinite dilution and then define a new activity coefficient, such that:

$$\gamma_i^* x_i H_i = \overline{f}_i^v$$

Chemical potential of solids dissolved in liquids

For liquid mixtures, we have defined chemical potential using:

$$\mu_i = \underline{G}_i + RT \ln \gamma_i x_i$$

$\underline{G}_i$ is the pure liquid at the same T,P as the mixture

This state does not exist for a dissolved solid

In order to cope with this problem we re-define the standard state

We use a standard state based on the infinite dilution limit of the solute, i.e. we define γ_i to be unity at infinite dilution

It is also common to use units of molality (mol kg^{-1})

Then:

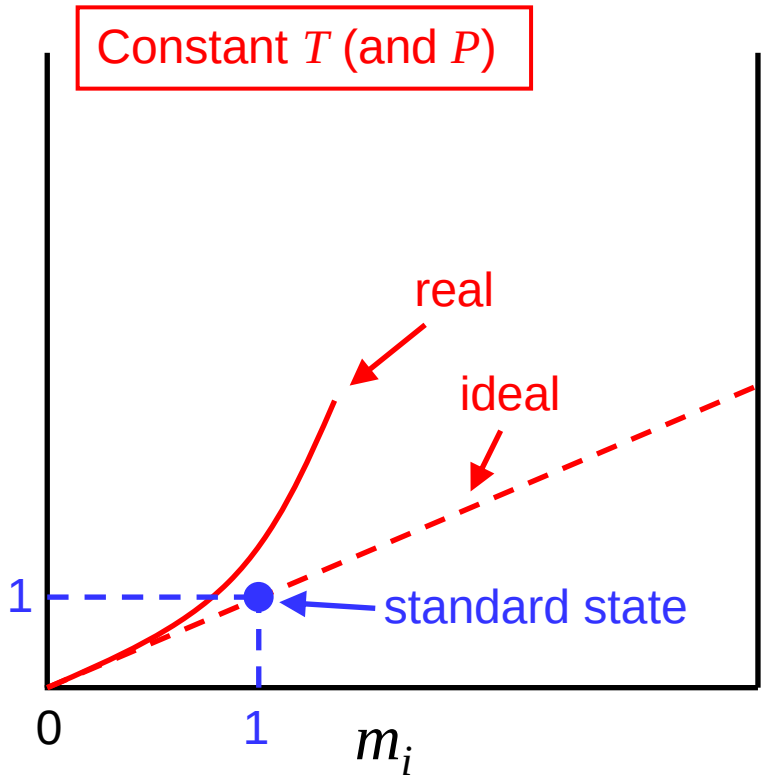
a new definition of activity coefficient

$$\mu_i = \mu_i^0 + RT \ln \gamma_i^{\#} m_i$$ ← strictly m_i/molkg^{-1}

chemical potential of "ideal" solute of unit molality

This is best thought about graphically:

$$a_i^{\#} = \gamma_i^{\#} m_i$$



Activity of ions in solution

We often need to cope with species which ionise in solution. We need to modify our chemical potential expression to deal with this.

Consider, for example, dissolved NaCl. We now have:

$$\mu_{\text{total}} = \mu^0_{\text{Na}^+} + \mu^0_{\text{Cl}^-} + RT \ln \gamma^{\#}_{\text{Na}^+} \frac{m_{\text{Na}^+}}{\text{molkg}^{-1}} + RT \ln \gamma^{\#}_{\text{Cl}^-} \frac{m_{\text{Cl}^-}}{\text{molkg}^{-1}}$$

We know that

$$m_{\text{Na}^+} = m_{\text{Cl}^-} = m$$

And we can define a **mean ionic activity coefficient** (since there is no way of measuring the individual ionic activities separately):

$$\gamma_{\pm} = \left(\gamma^{\#}_{\text{Na}^+} \gamma^{\#}_{\text{Cl}^-} \right)^{1/2}$$

We now have:

$$a_i^{\#} = m_i \gamma_i^{\#}$$

$$\mu_{\mathrm{NaCl}} = \mu^0_{\mathrm{NaCl}} + RT \ln a^{\#}_{\mathrm{NaCl}}$$

$(\mathrm{molkg^{-1}})^2$

hypothetical standard state when $a^{\#}_{\mathrm{NaCl}} = 1$ extrapolated from infinite dilution

where

$$\mu^0_{\mathrm{NaCl}} = \mu^0_{\mathrm{Na^+}} + \mu^0_{\mathrm{Cl^-}}$$

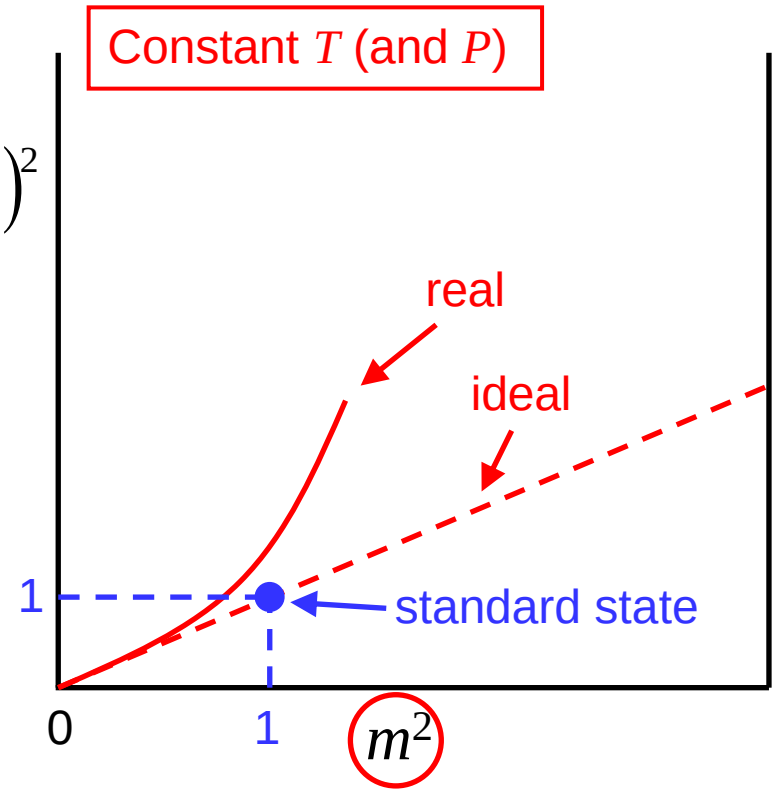
and

$$a^{\#}_{\mathrm{NaCl}} = (\gamma_{\pm} m)^2$$

$(\mathrm{molkg^{-1}})^2$

We do this because it makes no sense to define $\mu_{\mathrm{Cl^-}}$ and $\mu_{\mathrm{Na^+}}$ separately

$$a^{\#}_{\mathrm{NaCl}} = \left(\gamma^{\#}_{\mathrm{NaCl}} m\right)^2$$



For more complex electrolytes:

$$M_p A_q \rightarrow pM^+ + qA^-$$

$$a^{\#}_{\mathrm{M_pA_q}} = \left(a^{\#}_{\mathrm{M^+}}\right)^p \left(a^{\#}_{\mathrm{A^-}}\right)^q$$

and $$\gamma_{\pm}^{(p+q)} = \left(\gamma^{\#}_{\mathrm{M^+}}\right)^p \left(\gamma^{\#}_{\mathrm{A^-}}\right)^q$$

These changing definitions of activity coefficients

γ_i (pure liquid standard state)

γ_i^* (Henry's constant standard state)

$\gamma_i^{\#}$ (infinitely dilute dissolved solid extrapolated to unit molality standard state)

can be confusing

The key is to keep a firm grasp on the nature of the standard state you are using

Mean Ionic Activity Coefficient Models

Dissolved ionic solids are the hardest case

The Debye-Hückel limiting law is exact for infinitely dilute ionic solutions:

$$\ln \gamma_{\pm} = -\alpha \left| z_{+} z_{-} \right| \sqrt{I}$$

where α is a parameter depending on solvent and temperature

z_{+} and z_{-} are the ion valences

I is the ionic strength defined by $I = \frac{1}{2} \sum z_i^{\,2} m_i$

Unfortunately the Debye-Hückel limiting law can only be expected to be accurate for ionic strengths below 0.01 molal

For higher ionic strengths, there are a number of corrections to the Debye-Hückel limiting law available

It is beyond the scope of this course to go into details, but one example is:

$$\ln \gamma_{\pm} = -\frac{\alpha |z_+ z_-| \sqrt{I}}{1 + \beta a \sqrt{I}} + \delta I$$

Where β and δ are additional parameters

a is related to the average hydrated radius of the ions

Recall from lecture 10 part 3 (page 7 of handout):

$$\frac{dP}{dx_1} = (\gamma_1 P_1^{SAT} - \gamma_2 P_2^{SAT})(1 + x_1 \frac{d \ln \gamma_1}{dx_1}) = 0$$

Thus for a minimum or maximum in *P*, we have either

$$\gamma_1 P_1^{SAT} = \gamma_2 P_2^{SAT}$$ azeotrope

or $$\frac{d \ln \gamma_1}{dx_1} = -\frac{1}{x_1}$$ $\Rightarrow$ liquid-liquid separation (see later lecture)

Liquid-Liquid Equilibrium

All gases are mutually soluble at all compositions

This is not true for liquids

Some pairs of liquids mix perfectly
(e.g. toluene & benzene, ethanol & water, gin & tonic)
whereas others are immiscible
(e.g. n-heptane & water, carbon tet & water, olive oil & vinegar)

Partial miscibility is possible: at 70°C, β-picoline is soluble in water up to a mole fraction of ~11%
Beyond this concentration, the mixture separates into a water-rich phase ($x_{H_2O} \approx 89\%$) and a β-picoline-rich phase ($x_{\beta} \approx 58\%$)

In fact, there is always some degree of
mutual solubility even for “immiscible liquids”

Although often a nuisance, immiscibility does have its uses:

Examples

Steam distillation – essential oils (see VLLE later)

Polymerisation reactors – control of size by stirring

Liquid-liquid extraction – e.g. furfural, “the cornflakes chemical”, is used for refining lubricating oils from heavy oils (BP ~550°C) in oil refineries
Furfural was historically made from carbohydrates (pentosans) in corn cobs (or wood)
The result was a large surplus of crushed corn, which is said to have led to the invention of cornflakes

Preparation of microcapsules
– e.g. for slow release drugs

Remember from azeotropes lecture (handout 10)

$$\frac{dP}{dx_1} = \underbrace{\left(\gamma_1 P_1^{SAT} - \gamma_2 P_2^{SAT}\right)}_{\text{azeotrope}} \underbrace{\left(1 + x_1 \frac{d \ln \gamma_1}{dx_1}\right)}_{\text{phase separation}} = 0$$

$$\underline{G} = \underbrace{x_1 \underline{G}_1 + x_2 \underline{G}_2}_{\text{pure}} + \underbrace{x_1 RT \ln x_1 + x_2 RT \ln x_2}_{\text{ideal mixing}} + \underbrace{x_1 RT \ln \gamma_1 + x_2 RT \ln \gamma_2}_{\text{non-ideal mixing}}$$

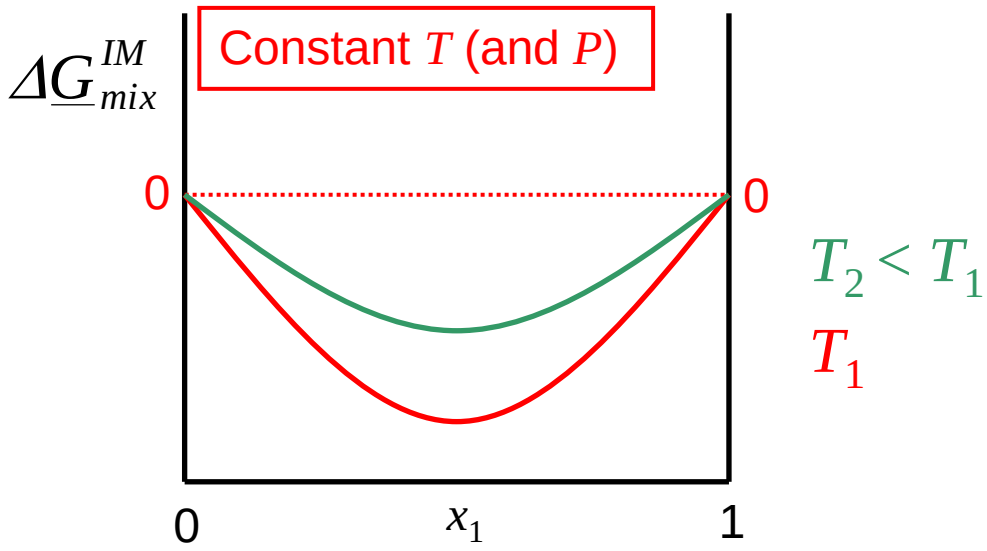
$$\frac{d^2 \underline{G}}{dx_1^{\,2}} = \frac{RT}{x_1 x_2} \underbrace{\left(1 + x_1 \frac{d \ln \gamma_1}{dx_1}\right)}_{\text{if this term is zero, then } \frac{d^2 \underline{G}}{dx_1^{\,2}} = 0}$$

We will now investigate the significance of points at which $\frac{d^2 \underline{G}}{dx_1^{\,2}} = 0$

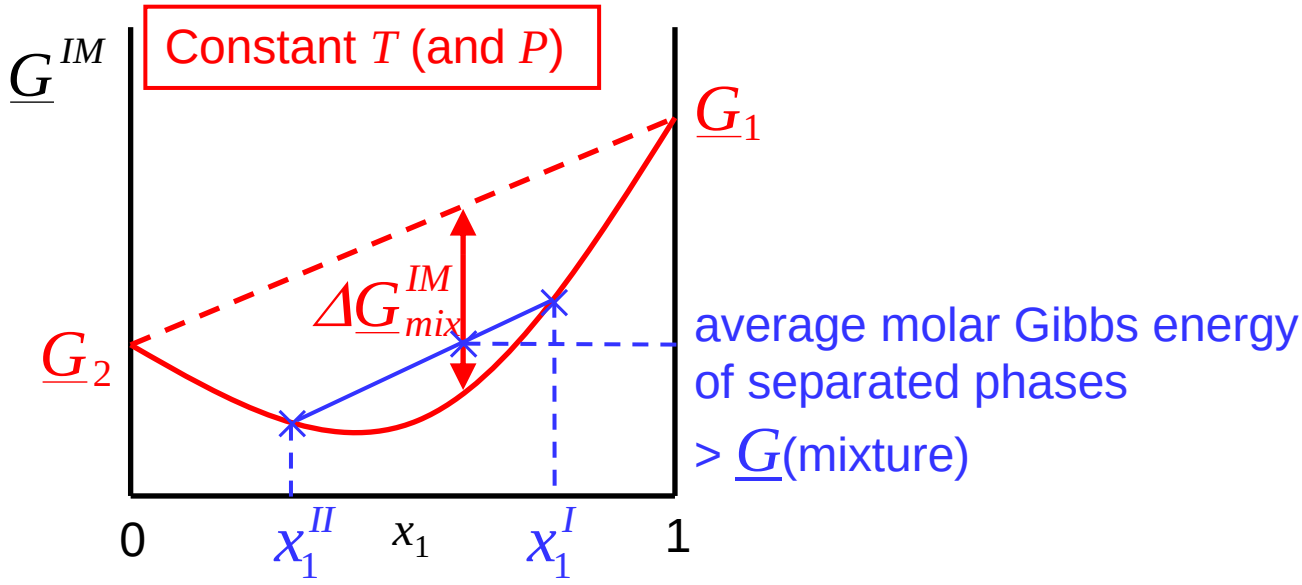
Recall that for an ideal, binary mixture:

$$\Delta \underline{G}_{mix}^{IM} = \mathrm{R}T\left(x_1 \ln x_1 + x_2 \ln x_2\right) < 0$$

Mixing is always thermodynamically favourable in the ideal case



$$\underline{G}^{IM} = x_1 \underline{G}_1 + x_2 \underline{G}_2 + \Delta \underline{G}^{IM}_{mix}$$



For a real mixture:

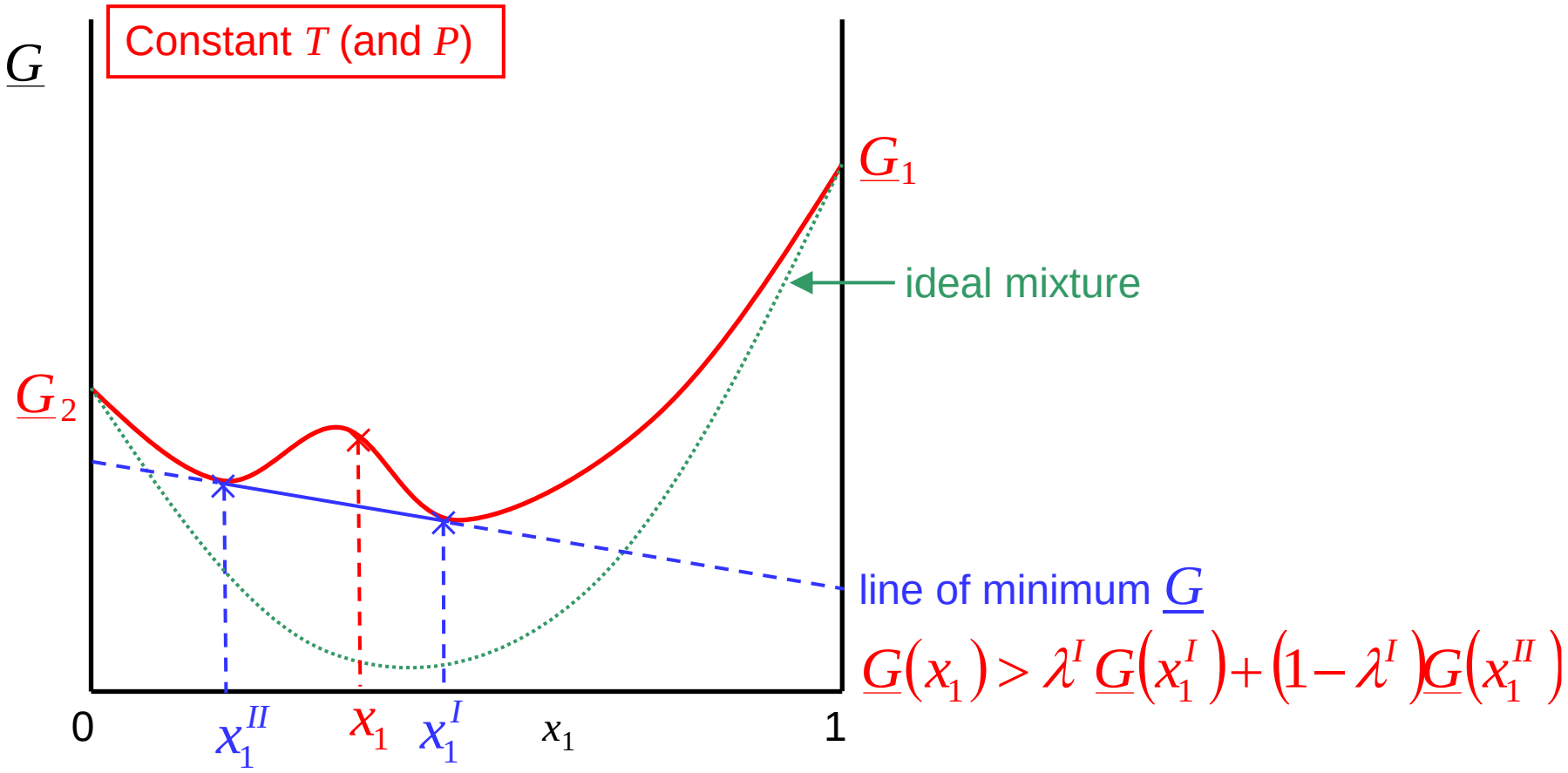
$$\Delta \underline{G}_{mix} = \mathrm{R}T\left(x_1 \ln x_1 + x_2 \ln x_2\right) + \underline{G}^{ex} \qquad \underline{G}^{ex} = \mathrm{R}T\left(x_1 \ln \gamma_1 + x_2 \ln \gamma_2\right)$$

e.g. $\underline{G}^{ex} = A x_1 x_2$ 1 constant Margules

where $\underline{G}^{ex} = \underline{H}^{ex} - T\underline{S}^{ex}$

If the unlike (1-2) interactions are repulsive by comparison with the like (1-1) and (2-2) interactions then the enthalpy change associated with mixing will be positive and may locally dominate over the entropy term

$\Rightarrow$ **phase separation**



Between $x_1 = x_1^I$ and $x_1 = x_1^{II}$ it is thermodynamically favourable for the homogenous mixture to split into two heterogeneous phases with compositions $(x_1^I, 1-x_1^I)$ and $(x_1^{II}, 1-x_1^{II})$

The criteria for equilibrium are:

$$\mu_1^I = \mu_1^{II} \quad \text{and} \quad \mu_2^I = \mu_2^{II}$$

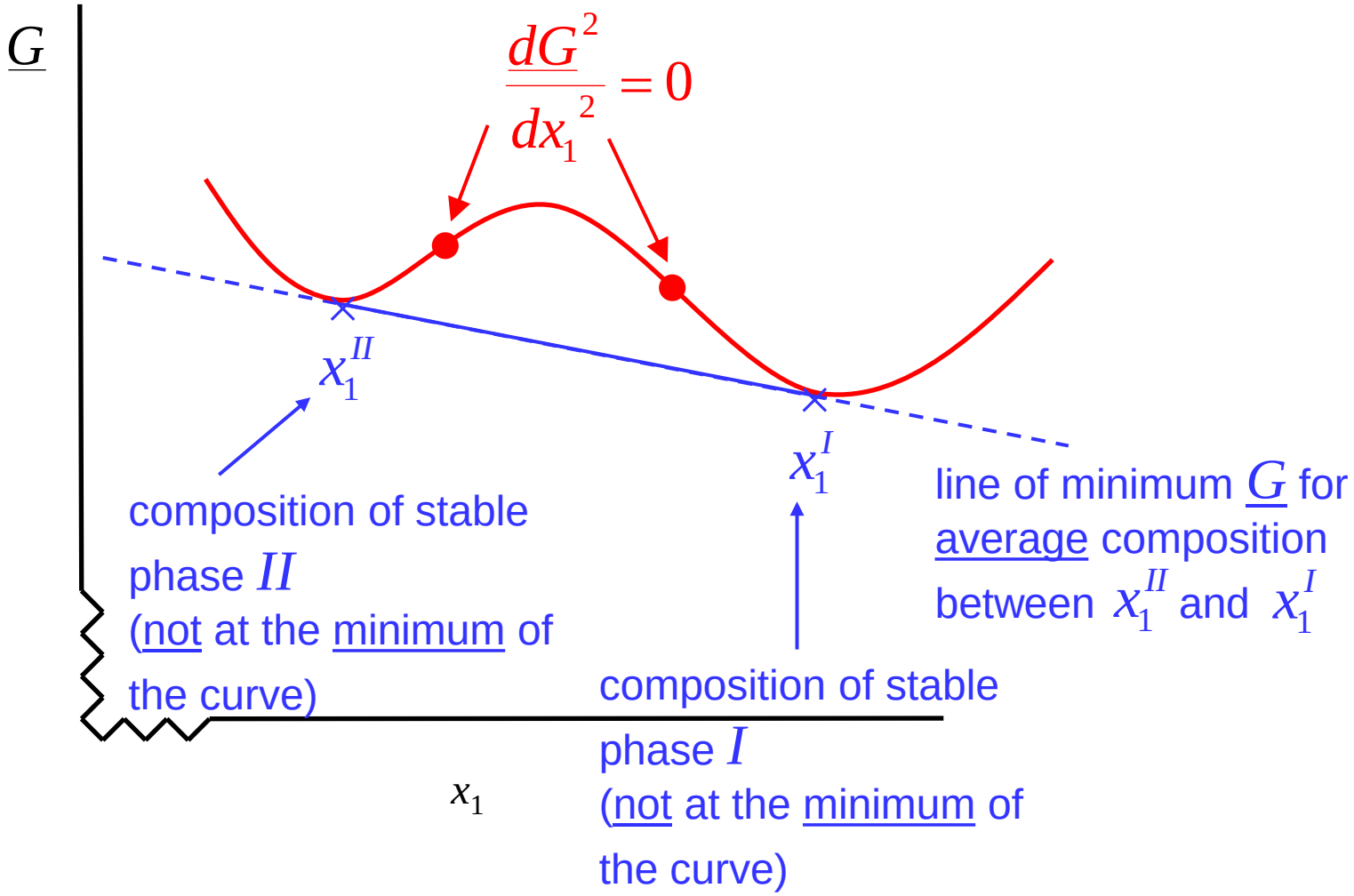
Now, $\underline{G} = x_1\mu_1 + x_2\mu_2$

so $$\frac{d\underline{G}}{dx_1} = \mu_1 - \mu_2 + x_1\frac{d\mu_1}{dx_1} + x_2\frac{d\mu_2}{dx_1}$$ G-D

$$= \mu_1 - \mu_2$$

$$\Rightarrow \quad \left.\frac{d\underline{G}}{dx_1}\right|_I = \mu_1^I - \mu_2^I = \mu_1^{II} - \mu_2^{II} = \left.\frac{d\underline{G}}{dx_1}\right|_{II}$$

i.e., the slope of the $\underline{G} - x_1$ curve is the same at the phase compositions x_1^I and x_1^{II}, as shown in the diagram below:
(detail from curve on previous page)



Condition for the existence of "wiggle" is that $\frac{dG^2}{dx_1^2} = 0$